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High-Resolution, Multiplex Antibody Patterning using Micropillar-Focused Droplet Printing, and Microcontact Printing

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Antibody arrays have great implications in many biomedical settings. However, commonly used patterning methods have difficulties in generating antibody arrays with both high resolution and multiplexity, limiting their applications. Here, a convenient and versatile technique for the patterning of multiple antibodies with resolution down to 20 µm is reported using micropillar-focused droplet printing and microcontact printing. Droplets of antibody solutions are first printed and stably confined on the micropillars of a stamp, and then the antibodies absorbed on the micropillars are contact-printed to the target substrate, generating antibody patterns faithfully replicating the micropillar array. The effect of different parameters on the patterning results is investigated, including hydrophobicity of the stamps, override time of the droplet printing, incubation time, and the diameters of the capillary tips and micropillars. To demonstrate the utility of the method, multiplex arrays of anti-EpCAM and anti-CD68 antibodies is generated to capture breast cancer cells and macrophages, respectively, on the same substrate, and successful capturing of individual cell types and enrichment among the cells are achieved. It is envision that this method would serve as a versatile and useful protein patterning tool for biomedical applications.

1. Introduction

Patterning of proteins, especially antibodies, on substrates has important biomedical applications. For example, antibody microarrays have been routinely used for proteomic analysis in biological research and clinical studies.[1] With spot sizes comparable to single cells (≈20 µm), antibody arrays have further found applications for the secretomic analysis of single cells.[2,3] In addition, micropatterned protein or antibody arrays have been utilized for cell anchoring to facilitate the study of cell migration,[4] proliferation,[5] mechanotransduction,[6] cell-cell interaction,[7] etc.

A commonly used method for the fabrication of protein microarrays is microarray spotting using commercialized spotters, which utilize the mechanism of inkjet printing or piezoelectric fluid dispensing.[8–10] Spotting techniques can generate multiplexed protein arrays by loading different protein solutions into the printing nozzles with high throughput and high automation.[11] Nevertheless, the dimension of the protein spot is highly dependent on parameters such as the rheological properties of the ink, evaporation, and the contact angle of the substrate, and the minimum spot is generally larger than 100 µm, making it less useful in applications on the single-cell level. In addition, the geometry of the protein spots is limited to a circular shape, making it nearly impossible to identify different spots without referring to spatial information.

In contrast, techniques such as microcontact printing (µCP),[12] dip-pen nanolithography (DPN),[13] and polymer pen lithography (PPL)[14] offer means to pattern proteins with resolutions down to sub-micro and even nanometer scopes. µCP uses elastic stamps with customized structures to transfer protein patterns to the final substrate. Since the fabrication of the elastic stamp is based on photolithography and soft lithography, which are standardized processes for the fabrication of microscale structures with high throughput, µCP has the ability to create patterns as fine as submicron-scale over a large area in a single stamping step. Expect for fulfilling a biologically demanded size regime of 10 to 100 µm, µCP enables flexible pattern designs with low cost and easy bench-top operations, spawning many applications in biomedical researchers. Despite that, the main drawback of µCP for biomedical applications lies in the difficulty to print different materials with one stamp. The generation of multiplex protein arrays using µCP requires multiple stamping with microscale alignment, which is difficult

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to operate, limiting its applications.[15] DPN utilized an atomic force microscope (AFM) cantilever as a writing "pen" to deposit "inks" to a substrate. Compared with μCP, DPN hold the advantage of its innate capacity for multiplexed depositions. However, to achieve large-area patterning, two-dimensional arrays of tips are required,[16] which are fragile and costly. And the process is time-consuming as well. The low throughput, lack of patterning flexibility, and requirement for costly devices are the main shortcomings of the techniques of DPN. PPL combined precise control of DPN over feature sizes and the intrinsic high throughput of μCP, utilizing elastomeric pyramidal tips as replacement of AFM cantilevers to transport "inks" to substrates. The sizes of the patterns created through PPL are force- and time-dependent, which can achieve a resolution down to 50 nm. Nevertheless, multiplexed patterning with PPL still requires alignment, which contains a series of complicated processes including the leveling of the stamp, offset correction, the coating of different sections of the stamp and so on.[17,18] Moreover, a costly and demanding DPN setup was generally employed for PPL. And the procedure to create large-scale patterns is significantly time-consuming and cumbersome compared with μCP, which is caused by a requirement for repeated ink reloading since the ink reservoir is limited to the surface of the tips.

Microchannel cantilever spotting (μCS), as another approach of scanning probe lithography (SPL) besides DPN and PPL, also has the advantage of high resolution since the utility of scanning probes allows SPL methods to bypass the optical diffraction limit, compared with photolithographic methods.[19] And its major comparative advantage over DPN and PPL is the embedment of a reservoir in the writing tip for liquid loading to overcome the ink reloading issue. A surface patterning tool (SPT) is designed with a reservoir, a split gap, and a cantilever shank. The ink was transferred from the reservoir through a microfluidic channel to the extremity of the cantilever with a split gap.[20] μCS can generate multiplexed arrays through a loading-spotting-washing-reloading cycle[21,22] or multiple-cantilever SPTs.[23] Despite the favorable aspects, compared with μCP, the size regime of μCS is limited to a relatively small range when the SPT design is given, normally 10 to 40 μm. Larger feature sizes in the range of 40 μm to 100 μm, which are also frequently required in biomedical experiments, are difficult to address with the specific SPT. And as a direct spotting method, the geometry of the patterns is also fixed to circular shapes, whereas the shape of patterns can be a useful tool for coding different "inks" in the same array. In addition, like other SPL approaches, DPN setup is still required and additional SPT pens are also needed, pushing up the cost of devices.

Though other works have also been proposed to generate protein patterns with high resolution using methods such as photolithography,[24] microfluidics,[3,25] dip-pen nanolithography,[26] and surface wettability,[27] the throughput or multiplexity is still suboptimal. In this work, we developed a convenient and versatile technique for the patterning of multiple antibodies with high resolution using micropillar-focused droplet printing and microcontact printing. Protein solutions were first printed on each micropillar of a stamp, before the absorbed protein patterns were contact-printed to the target substrates. The micropillars on the stamp confined the printed droplet to its tip, thus enabling the generation of protein patterns with shapes and resolutions defined by the micropatterns of the stamp. In addition, multiplex protein arrays could be generated by spotting each micropillar with the designated protein solution in a highly automated manner. We showed that this technique could generate multiplex antibody arrays with spot diameters as small as 20 μm. We systematically investigated the parameters that influence the quality of the printed protein patterns, namely the hydrophobicity of the stamp, override time of the spotting, incubation time, and the nozzle diameter. To demonstrate the utility of this technique, we generated arrays of two antibodies targeting the surface markers of breast cancer cells and macrophage cells, respectively, and successfully enriched the two types of cells on different spots, showing the potential application in the study of cell-cell interactions such as paracrine by localizing colonies of different cells in proximity. Compared to a microarray spotter, this technique holds the advantages of higher resolution and easier control over the shapes and sizes of the resultant spots. Compared to SPL methods, the superiorities of this technique lie in the overall simple settings, high throughput, and the easy patterning process. We envision that the multiplex, versatile antibody arrays generated by this technique would be useful in biomedical applications involving multiple cell capturing or multiple analyte detection.

2. Results

2.1. Overview of the Antibody Patterning Method

The antibody patterning method leveraged the advantages of both droplet printing and microcontact printing (μCP) to generate high-resolution, multiplex antibody arrays. The patterning process started with printing droplets of antibody solution using droplet printing onto the tips of micropillars of PDMS stamps, as shown in **Figure 1**a. Due to surface tension, the antibody droplet was confined to the micropillar tip without spreading over (Figure 1b), effectively localizing the droplet on the target micropillar. In addition, multiple antibody printing was readily enabled in this process by switching inks. The droplets evaporated quickly and formed a precipitate on the micropillar tips, as shown in Figure 1c, preventing the antibody absorption onto the micropillar tips, which was mainly induced by the Van der Waals force.[28] To address that, we placed the stamp in a humid, warm chamber and incubated it for a few hours to rehydrate the antibody and facilitate the adsorption onto the micropillar tips. The stamps were then washed and blow-dried, and μCP was performed by placing the stamps onto surface-activated coverslips, generating antibody arrays with well-defined shapes, as shown in Figure 1d. We further used atomic force microscopy to characterize the surface topology of the antibody spots, and the results showed that the thickness was roughly 10 nm (Figure 1e). This method was able to generate antibody arrays with spot sizes as small as 20 μm with clear-cut spot boundaries. (Figure 1f). In addition, spots with various shapes, such as circles, rectangles, and triangles, could be generated by using stamps with different designs (Figure 1g). This merit could provide a means to encode the antibodies with a single fluorescent channel by printing droplets of different antibody solutions on the tips of micropillars in different shapes. Using inks of different antibodies, this method further enabled the generation of multiplex antibody arrays with spot diameters

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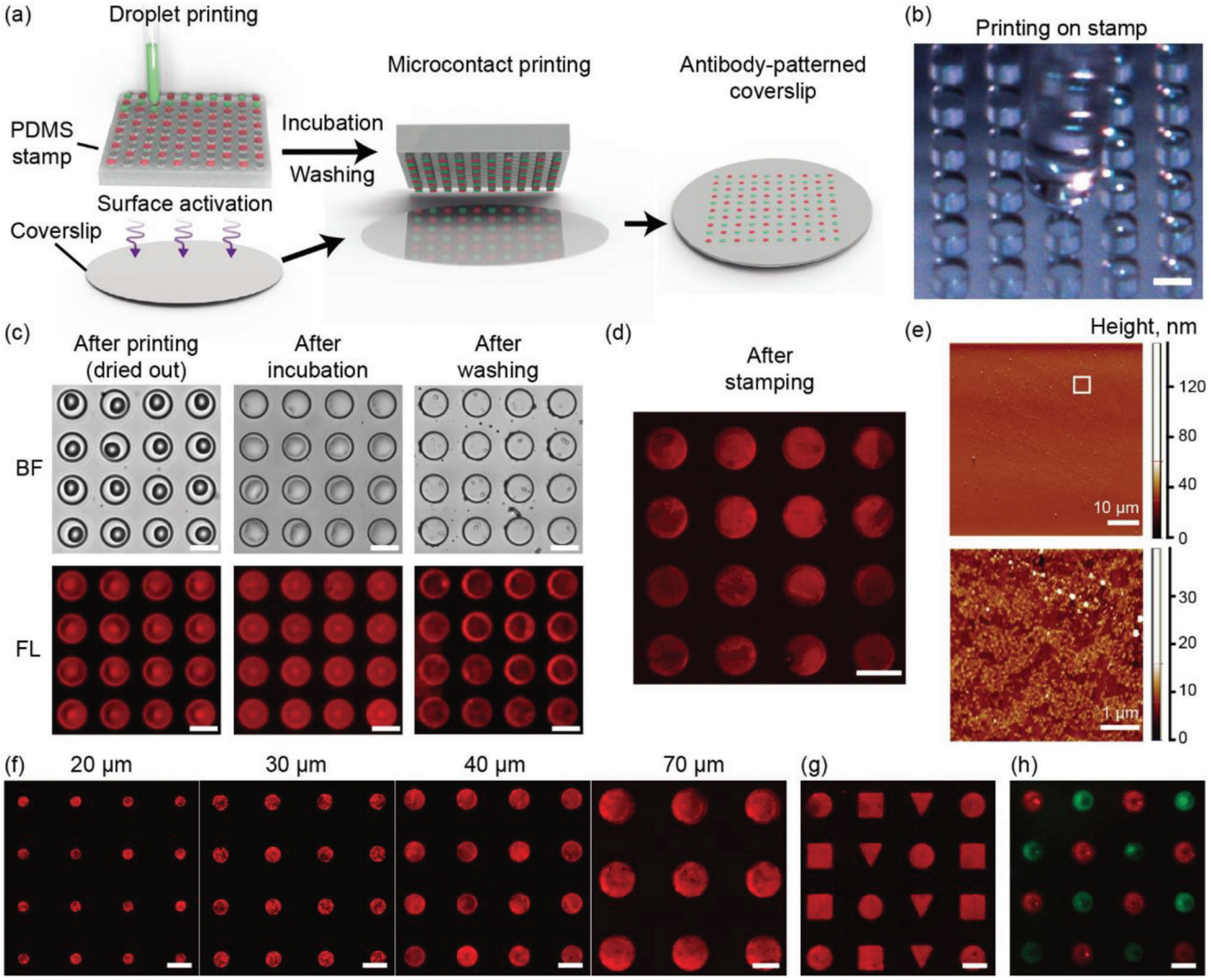

Figure 1. Overview of the antibody patterning method. a) Schematic showing the antibody patterning process. Antibody droplets are printed on the micropillars of a PDMS stamp, and the absorbed antibodies are contact-printed to a surface-activated coverslip. The micropillars confined the printed antibody droplet on its tip, thus enabling high resolution and easy control over the shape of the resultant antibody arrays. b) Micrograph showing the process of the droplet printing on the stamp. Scale bar, 50 μm. c) Micrographs showing the PDMS stamps after droplet printing, after incubation, and after washing. BF, brightfield. FL, fluorescence. Scale bars, 50 μm. d) Fluorescence micrograph showing the printed antibody array on the coverslip. Scale bar, 50 μm. e) Atomic force microscopy image showing the topography of the printed antibody spots on the coverslip. Zoom-in view of the boxed area in the upper panel is shown in the lower panel. f) Fluorescence micrographs of the printed antibody arrays with different spot diameters as indicated. Scale bars, 50 μm. g) Fluorescence micrographs of the printed antibody arrays with different shapes. Scale bar, 50 μm. h) Fluorescence micrographs of the printed antibody arrays of two types of antibodies. Scale bar, 20 μm.

down to 20 μm, which was very difficult to achieve using microarray spotter or microcontact printing alone (Figure 1h).

2.2. Characterization of the Patterning Process

We then sought to characterize the patterning process and investigate how different parameters influenced the patterning results. Parameters that could have a major effect on the patterning results included the surface hydrophobicity of the stamp, the override time of the capillary tip on the stamp, the incubation time, the diameter of the capillary tip, and the diameter of the micropillar on the stamp.

The surface hydrophobicity of the PDMS stamps had two competing effects on the antibody patterning. While hydrophilic surfaces promoted the wetting of the antibody droplet on the micropillar tip and thus made the droplet printing easier, the associated high surface energy made it difficult to transfer the antibody from the stamp to the coverslip in the subsequent μCP process. In addition, an extremely hydrophilic surface could make the solution droplet climb down the micropillar and induce cross-contamination. Therefore, we performed experiments to determine the appropriate surface properties of the PDMS stamp.

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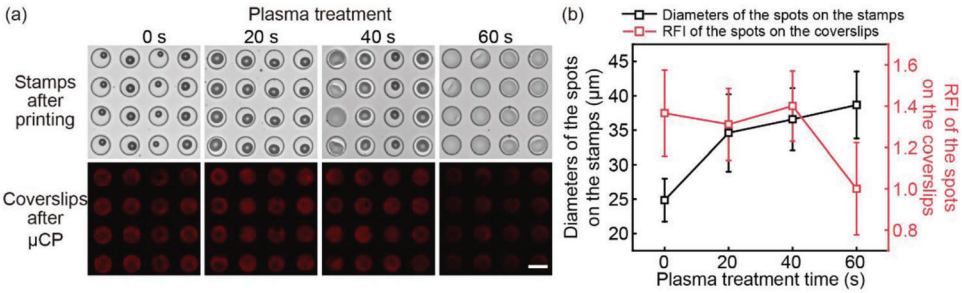

Figure 2. Effect of the hydrophobicity of the PDMS stamps on the patterning results. a) Representative micrographs showing the stamps after printing and the coverslips after μCP using PDMS stamps treated with oxygen plasma of different durations as indicated. Scale bar, 50 μm. b) Diameters of the spots on the stamps and the RFI of the spots on the coverslips as a function of the plasma treatment time. Data represent mean ± SD with n = 50 μCP, microcontact printing. RFI, relative fluorescence intensity.

We tuned the hydrophobicity of the PDMS stamps by treating the stamp surface with oxygen plasma for different durations. For example, after 40 s of treatment of oxygen plasma, the contact angle of the PDMS surface reduced from 102.1 ± 2.2° to 54.0 ± 1.4° (Figure S1, Supporting Information). We examined the patterning results using different plasma treatment times, namely 0, 20, 40, and 60 seconds. As anticipated, as the plasma treatment time increased, the diameters of the antibody spots on the stamps increased due to the improved surface wetting (**Figure 2**a). For example, when using PDMS stamps without plasma treatment, the diameter of the antibody spots on the stamps was 24.9 ± 3.1 μm, as shown in Figure 2b. In contrast, with a plasma treatment for 20 seconds, the diameter of the antibody spots on the stamps was 34.6 ± 5.6 μm, which was significantly larger. When plasma treatment time was higher than 20 s, the changes in spot diameters were no longer significant. In the meanwhile, we characterized the antibody amount transferred to the coverslips by measuring the fluorescence intensities. As shown in Figure 2b, the fluorescence intensity of the antibody spots on the coverslips remained relatively constant when the plasma treatment time was less than 40 s. When the treatment time increased to 60 s, the relative fluorescence intensity (RFI) decreased to 1.00 ± 0.22, which was significantly smaller than the RFI of 1.40 ± 0.17 when the plasma treatment time was 40 s, indicating that excessive plasma treatment could reduce the efficiency of antibody transferring from the stamp to the coverslip likely due to the elevated surface energy of the stamp. These experimental results suggested that plasma treatment could indeed facilitate the printing of the antibody solution onto the stamps, but the dosage should not exceed 40 s.

The override time of the capillary tip on the stamp surface determined the duration of the contact between the ink meniscus and the micropillar top before droplet formation, thus determining the wetting of the resultant droplet on the micropillar top. We anticipated that prolonged override time could lead to more even spreading of the antibody droplet on the stamps and consequently more efficient antibody transferring to the coverslips. We performed the patterning process with override times ranging from 0 to 2.5 s. We observed that as the override time increased, the antibody spots printed on the stamps intended to be more spreading and thus bigger, and the fluorescence of the antibody spots generated on the coverslips became brighter, as shown in **Figure 3**a. Quantified results showed that the diameters of the spots on the stamps and the fluorescence intensity of

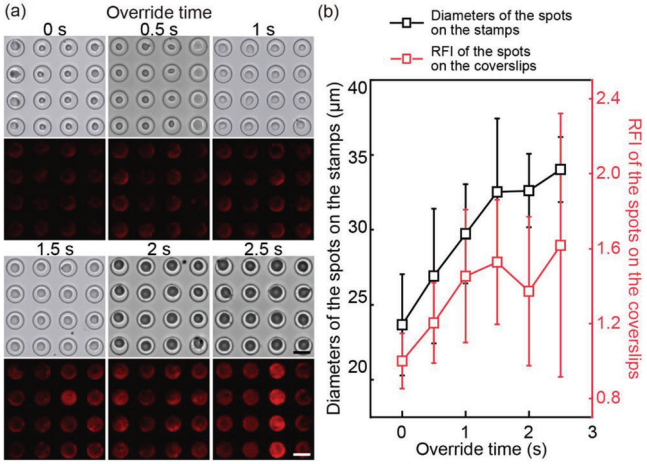

Figure 3. Effect of the override time of the droplet printing on the patterning results. a) Representative micrographs showing the stamps after printing and the coverslips after μCP using different override times. Scale bar, 50 μm. b) Diameters of the spots on the stamps and the RFI of the spots on the coverslips as a function of the override times. Data represent mean ± SD with n = 50.

the spots on the coverslips increased steadily before the override time reached 1.5 s, after which both the diameters of the spots on the stamps and the fluorescence intensity of the spots on the coverslips remained relatively unchanged (Figure 3b). These results suggested that longer override times could indeed facilitate droplet wetting on the pillars, providing a larger interface for antibody adhesion onto the stamp and resulting in more efficient transferring of antibodies to the recipient substrates, though the effect became negligible with override time larger than 1.5 s.

Absorption of the antibody onto the micropillar tops required an aqueous environment. After the droplet printing, the droplets on the stamps evaporated within a minute due to the small size and formed antibody precipitate on the micropillar tops, interrupting the absorption process. To address that, we incubated the stamps in a humid chamber to moisturize micropillars and re-dissolve the antibody. As a result, the coverslips received significantly more antibodies after μCP compared to that without incubation (Figure S2, Supporting Information). We then sought to investigate how the incubation time affected the patterning

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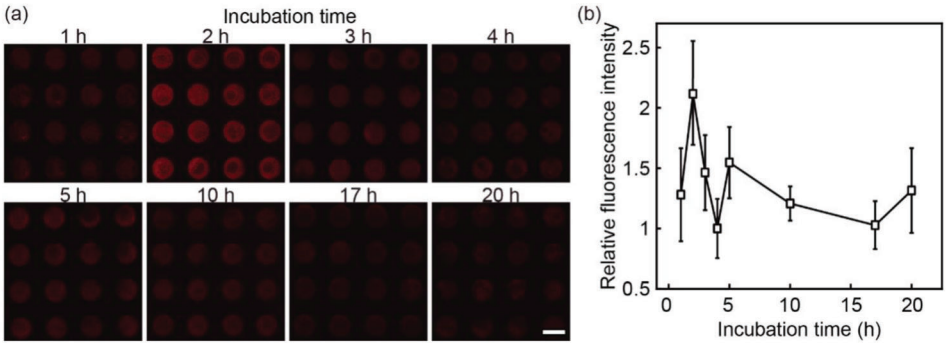

Figure 4. Effect of the incubation time on the patterning results. a) Representative micrographs of the coverslips after μCP using different incubation times. Scale bar, 50 μm. b) Relative fluorescence intensity of the antibody arrays as a function of the incubation times. Data represent mean ± SD with n = 50.

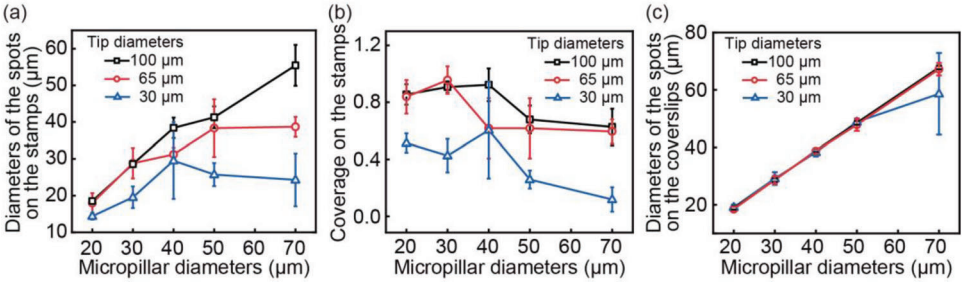

Figure 5. Effect of the micropillar diameters and the tip diameters on the patterning results, including a) diameters of the spots on the stamps, b) coverage on the stamps, and c) diameters of the spots on the coverslips as a function of the micropillar diameters using different tip diameters as indicated. Data represent mean ± SD with n = 100.

results. We incubated the stamps for different durations, namely 1, 2, 3, 4, 5, 10, 17, and 20 h, and performed the patterning process. As shown in **Figure 4**, two hours' incubation obtained the optimal patterning effect, with the RFI of the antibody spots being 2.14 ± 0.43, which was significantly greater than the 1.29 ± 0.40 at 1 hour's incubation, the 1.48 ± 0.32 at 3 hour's incubation, and other experimental conditions. This observation suggested that an incubation time of 2 h should be adopted.

The sizes of the dispensing capillary tips and the micropillars could also have a major impact on the patterning results. To investigate that, we performed the patterning process using capillary tips with different inner diameters, namely 30, 65, and 100 μm, and stamps with micropillars of different diameters, namely 20, 30, 40, 50, and 70 μm, and characterized the diameters of the spots on the stamps, the coverage on the stamps, and the diameters of the spots on the coverslips. The results showed that when the micropillar diameters were smaller than the tip diameters, the diameters of the spots on the stamps increased as the micropillar diameters increased and the coverage remained relatively unchanged (**Figure 5**a,b). When the micropillar diameters were larger than the tip diameters, the contact was mainly limited by the relatively small tips (Figure S3, Supporting Information). Consequently, as the micropillar diameters increased, the spot diameters did not appear to change, and the coverage on the stamps declined (Figure 5a,b). Nevertheless, the antibody spots on the coverslips after μCP showed great consistency as the shape of the micropillars with similar diameters, as shown in Figure 5c. For example, when using capillary tips with diameters of 65 μm, the diameter of the antibody spots on the coverslips were 18.5 ± 0.9, 28.7 ± 1.4, 38.7 ± 0.9, 47.8 ± 2.2, and 67.2 ± 2.2 μm for micropillars with diameters of 20, 30, 40, 50, and 70 μm, respectively. When 30 μm capillary tips and 100 μm micropillars were used together, the antibody spots were slightly smaller than the designed shape, likely due to the insufficient spreading of the antibody droplet on the micropillars. Therefore, small capillary tips should not be used along with micropillars with much larger diameters. These results demonstrated the capability of this patterning method in generating high-resolution antibody arrays with predefined geometries. In addition, patterns finer than the capillary tips can be generated by using stamps with micropillars smaller than the capillary tips.

2.3. Cell Patterning using Antibody Arrays

To demonstrate the utility of this patterning method, we sought to pattern cells using the generated antibody arrays. Given the importance of cancer microenvironments and the interaction between cancer cells and immune cells, we chose breast cancer cells (MCF-7) and macrophages for this study. MCF-7 cells express the surface marker of epithelial cell adhesion molecule (EpCAM), and human macrophages, which were differentiated

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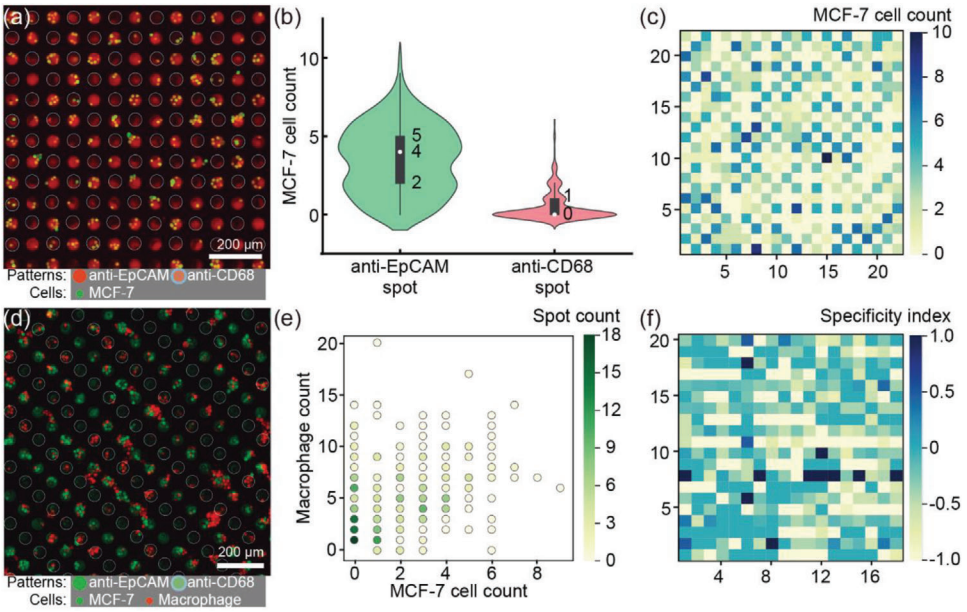

Figure 6. a–c) Capturing of MCF-7 (EpCAM+) using antibody arrays with alternating anti-EpCAM and anti-CD 68 antibodies. a) Fluorescence micrograph showing the captured MCF-7 cells (green) and the antibody arrays (red) of anti-EpCAM antibody (bright red) and anti-CD68 antibody (dim red and false-circled in blue). b) Violin plot showing the distribution of the number of MCF-7 cells on spots of anti-EpCAM antibody and anti-CD68 antibody, respectively. c) Heat map of the MCF-7 cell counts on each spot of the antibody arrays. d–f) Co-capturing of MCF-7 (EpCAM+) and macrophages (CD68+) using antibody arrays with alternating columns of anti-EpCAM and anti-CD 68 antibodies. d) Fluorescence micrograph showing MCF-7 cells (green), macrophage cells (red), and the antibody arrays (green) of anti-EpCAM antibody (bright green) and anti-CD68 antibody (dim green and false-circled in blue). e) Scatterplot showing spots with different cell counts. f) Heat map showing specificity index of cell patterns. The specificity index is defined as the ratio of the difference between MCF-7 count and macrophage count to the total cell count.

from a human monocyte cell line (THP-1), express the surface marker of the cluster of differentiation 68 (CD68). Therefore, we fabricated antibody arrays with both anti-EpCAM and anti-CD68 antibody spots, aiming to specifically capture MCF-7 cells and macrophages, respectively.

We first aimed to characterize the capturing efficiency and specificity by seeding MCF-7 cells alone. We stained all the antibody spots with a single fluorescing secondary antibody, and the fluorescence micrograph showed alternating intensity across the spots, indicating successful patterning of the two antibodies (Figure S4, Supporting Information). We also observed successful cell capturing onto the antibody spots, and anti-EpCAM spots generally captured more MCF-7 cells than anti-CD68 spots, as shown in **Figure 6**a. 90.5% of the anti-EpCAM spots successfully captured MCF-7 cells, and 79.2% of the anti-CD68 spots showed no MCF-7 cell adhesion. We further counted the cells present on each spot, the results showed that the average cell count on anti-EpCAM spots was 3.48 ± 2.17, with the majority of the spots capturing 3∼4 cells (Figure 6b). The average cell count on the anti-CD68 spots was 0.48 ± 0.96. We then generated a heatmap of the cell count on each spot to visualize the cell distribution across the antibody array, as shown in Figure 6c. The heatmap showed alternating intensity, with spots of high and low cell count well dispersed, indicating that the specific capturing of MCF-7 was successful.

We then sought to pattern the two types of cells on the antibody array. We first seeded MCF-7 cells (EpCAM+) onto the coverslip with the antibody array and washed away the unadhered cells. We then seeded the differentiated macrophages (CD68+) on the coverslip. As shown in Figure 6d, MCF-7 cells were mostly confined to spots with anti-EpCAM antibodies, and spots with anti-CD68 antibodies captured mostly macrophages, indicating that specific cell capturing using the antibody arrays was technically feasible. However, due to the inherent adhesion between macrophages and MCF-7 cells, we observed the appearance of macrophage cells on the spots with anti-EpCAM antibodies (Figure 6d). We also investigated the approach of incubating macrophages before seeding MCF-7 cells, while obtaining undesirable outcomes (Figure S5, Supporting Information). When seeding macrophages first, it adhered to both anti-EpCAM and anti-CD68 spots. A potential reason was that macrophages held an inherent ability tononspecifically adhering to a large variety of surfaces.[29] Thus the procedure of cell patterning was completed as we introduced above. We counted the quantity of each type of cell on each spot and consolidated the data based on cell counts, showing that the spots were mainly concentrated in two regions (Figure 6e). The first region included spots that captured macrophage cells but with almost no MCF-7 cells, which were most likely the spots of anti-CD68 antibodies. The other region included spots with the presence of both MCF-7 cells and macrophages, which were likely the spots of anti-EpCAM antibodies and captured MCF-7 cells first before macrophages adhered. To further visualize the cell pattern, we defined specificity index as the ratio of the difference between MCF-7 count and macrophage count to the total cell count, with 1.0 representing the capturing of solely MCF-7 cells and −1.0 representing the capturing of solely macrophage cells. We then plotted the heatmap to visualize the spatial distribution of the specificity index of the spots, as shown in Figure 6f.

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The plot showed alternating patterns in the vertical direction and long stripes of negative specificity index which represented macrophage capturing, though the pattern did not appear as striking as anticipated due to the cell-cell adhesion. These results of cell patterning using multiplex antibody arrays demonstrated the potential scientific applications of this patterning method.

3. Discussion and Conclusions

In this work, we developed a method for protein patterning with high resolution and multiplexity. Compared to the μCP technique which is difficult to pattern multiple proteins, this method leveraged the advantages of both piezoelectric printing and microcontact printing to realize high-resolution multiplexed protein arrays. Compared to direct spotting methods, including DPN and μCS, this method did not require costly DPN setup to achieve size regime of 20 to 100 μm that is relevant for many biomedical applications. This method also held the advantages of simple printing operation and high throughput of patterning. Compared to PPL and μCP, multiplexed patterning of this method is simpler, more straightforward, and more flexible. In addition, as validated by the experiments, the focusing effect provided by the micropillars ensured the robustness and reliability of the patterning. Therefore, with the merits of high resolution, reliability, and throughput, this method could be a useful tool in various applications.

Currently, the resolution of this method is ≈5 μm, which is mainly limited by the resolution of the digital camera coupled with the spotter. By adopting cameras with higher magnification and resolution, the printing resolution can be further improved. In addition, an inherent limitation of microcontact printing is the compromised immunological function of the antibody, presumably due to inadequate epitope exposure after the microcontact printing.[30] Future work shall be devoted to improving the immunological function of the generated antibody array. For example, by combining indirect protein patterning methods, such as self-assembly monolayer[31] or oligonucleotide coding,[25] the function of the antibodies on the arrays could be improved, potentially enabling more applications such as multiplex detection of single-cell secretion. For example, in order to detect the secretion of single cells, microwells could be bonded onto substrates with antibody microarrays to trap single cells in the microwells and capture the secretion of the cells with the antibodies. Printing in a humid environment or using an aqueous two-phase system can also potentially provide better antibody function.

4. Experimental Section

Fabrication of the PDMS Stamps: The PDMS stamps were fabricated by soft lithography using SU-8 molds. Fabrication of the SU-8 molds was outsourced to a microfabrication company (Suzhou Research Materials Microtech Company, China) and generated using photolithography. The SU-8 molds were then treated with oxygen plasma (PDC-002, Harrick Plasma) for one minute and placed in a vacuum chamber with silane vapor (448 931, Sigma-Aldrich) for 10 hours. PDMS prepolymer (base:curing agent, 20:1 w/w; Sylgard 184, Dow, Inc.) was then poured onto the SU-8 molds before being baked at 60 °C for 10 h to fully cure. Compared to the commonly used base-to-curing agent ratio of 10:1, the ratio of 20:1 offered more conformal contact. The cured PDMS was then peeled off, cut into desired shapes, and cleaned using ethanol. When the hydrophobicity needed tuning, the PDMS stamps were treated with oxygen plasma for prescribed durations, typically 40 s.

Protein Patterning: Droplet printing was performed using an ultrasonic fluid dispensing instrument (Microplotter Proto, SonoPlot, Inc.). Glass capillaries with different tip diameters were fabricated using a pipette puller (P-97, Sutter Instrument, Inc.). Unless otherwise mentioned, tip diameters of 65 μm were used. Patterns of the antibody arrays were designed in the SonoDraw software (SonoPlot, Inc.) and uploaded to the control software for spotting. The antibody solutions for the inks were all diluted in PBS (BL551A, Biosharp Life Sciences) to 50 μg mL^{-1}. Override times of the capillary tip on the micropillar tip were typically set at 1.5 s and tuned as needed. The dispensing strength was kept constant at 1.2 V. When multiple antibody solutions were printed, the capillary was rinsed for three cycles in DI water before loading different inks. After printing on the stamps, the antibody spots dried. To rehydrate the spots and facilitate absorption, the stamps were incubated in a humid, warm chamber (100% relative humidity, 37 °C) for typically 2 h. The stamps were then rinsed with PBS and blow-dried. Microcontact printing was then performed to transfer the antibody patterns from the PDMS stamps to coverslips. The coverslips (0344-1820, Citotest Labware) were washed in ethanol, blow-dried, and treated with UV ozone (BZS250GF-TC, Huiwo Technology). The stamps were then brought to contact with the coverslips and sit for one minute before being removed. The coverslips were then rinsed in PBS.

In the experiments of method characterization, IgG conjugated with Alexa Fluor 488 (A32731, Invitrogen), IgG conjugated with Alexa Fluor 568 (A11004, Invitrogen), and IgG conjugated with TRITC (HA1016, HuaBio) were used as the inks. In the experiments of cell capturing, recombinant monoclonal antibodies targeting cell surfaces markers, namely anti-EpCAM antibody (ab223582, Abcam) and anti-CD68 antibody (ab213363, Abcam), were used as the inks. In addition, the coverslips were pre-coated with PDMS to minimize non-specific cell adhesion. Coverslips were spin-coated with PDMS prepolymer (base:curing agent, 10:1 w/w) at 3000 rpm for 45 s using a spinner (EZ6-S, Lebo Science), resulting in an estimated thickness of 10 μm, and baked at 60 °C for 10 h, before being used for microcontact printing.

Cell Culture and Seeding: The MCF-7 cell line was cultured in DMEM (10 569 010, Gibco) containing 10% fetal bovine serum (FBS; 10270-106, Gibco) and 1% penicillin-streptomycin (P/S; 15140-122, Gibco). The THP-1 cell line was cultured in RPMI-1640 (11 875 093, Gibco) containing 10% FBS and 1% P/S. Cells were cultured in a cell culture incubator (IP610, Yamato Scientific) maintained at 37 °C with 5% CO_2. Differentiation of THP-1 cells to macrophages was achieved by exposure to phorbol-12-myristate 13-acetate (PMA; P8139, Sigma-Aldrich). THP-1 cells were seeded at a density of 18k cells cm^{-2} in a cell culture dish containing a cell culture medium supplemented with 0.05 ng ml^{-1} PMA. After 24 h, PMA was removed by replacing the cell culture medium, and the cells were cultured for another 72 h.

Before seeding cells on the patterned coverslips, the coverslips were immersed in 0.2% Pluronics F127 (P2443, Sigma-Aldrich) for one hour to prevent unspecific cell adhesion, rinsed with PBS, and transferred to a 12-well plate (150 628, ThermoFisher Scientific). Cells were seeded at a density of 53k cells cm^{-1},[2] and after 2 h of sitting, the coverslips were rinsed with PBS to remove cells that have not adhered. In the experiments of multiple cell patterning, MCF-7 cells were first seeded and washed before the differentiated macrophages were seeded at a density of 53k cells cm^{-1}.[2] After 2 h, the coverslips were washed to remove macrophages that have not adhered.

Imaging and Analysis: Atomic force microscopy (NTEGRA, NT-MDT LLC) was performed to characterize the surface topography of the antibody spots. Live cells were stained with CellTracker Red CMTPX (C34552, Invitrogen) and CellTracker Green CMFDA (C7075, Invitrogen) following the supplier's protocols. After cell-capturing experiments, antibody patterns were immunostained for visualization. The cells on the coverslips were fixed in 4% Paraformaldehyde (P0099, Beyotime Biotechnology) for one hour and washed with PBS several times. The coverslips were then blocked using 5% bovine serum albumin (A104912, Aladdin) for 30 min at room temperature. Subsequently, the coverslips were immersed in 2 μg mL^{-1}

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secondary antibodies (F2765 and A11004, Invitrogen) and incubated for 30 min, before being rinsed with PBS.

Micrographs were captured using a fluorescence microscope (Eclipse Ti2-E, Nikon) coupled with a CMOS camera (DS-Qi2, Nikon). Imaging settings, including excitation intensity, exposure time, and contrast adjustment were kept consistent when performing the comparison.

In the experiments of MCF-7 cell capturing, 264 anti-EpCAM spots, and 265 anti-CD68 spots were included for analysis. In the multiple-cell capturing, 173 anti-EpCAM spots and 129 anti-CD68 spots were included for analysis. Empty spots with no adhered cells were excluded. The numbers of cells within each antibody spot were manually counted.

To measure the contact angles of stamp surfaces treated with oxygen plasma of different doses, a drop of deionized water was pipetted onto the PDMS surface treated with oxygen plasma and imaged with a stereomicroscope (AE2000, Motic). The contact angles were then measured in ImageJ, which were defined as the angles between the solid-liquid interfaces and the tangent planes. Three droplets were imaged and analyzed for each duration of oxygen-plasma treatment.

Statistical Analysis: Images were analyzed in ImageJ (National Institute of Health, USA). Relative fluorescence intensity was calculated as the ratio of the average grayscale value of each pattern to the average grayscale value in one of the experiment groups, typically the one with the lowest value. All data represent mean ± SD and the sample sizes were declared in the captions of the figures. The two-sided student's t-test was used to access the differences between groups with a 95% confidence interval. Statistical analysis was carried out using SPSS software (International Business Machines Corporation, USA).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author contributions

M.J. and Z.L. conceptualized the project. M.J. and K.W. performed the characterization experiments. M.J., M.W., and Y.Z. designed and performed the cell experiments. M.J., C.Y., and Z.L. analyzed the data. M.J. and Z.L. wrote the manuscript. All authors contributed to the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

antibody microarray, cell patterning, droplet printing, microcontact printing, protein patterning

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